

SV & UVM Practice Questions

Part 8: Technical Interview Preparation

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1. What is UVM and the difference between SV and UVM?
2. What is a Regression testing?
3. Draw the block diagram of UVM or OVM?
4. Explain UVM architecture.
5. What are TLM ports?
6. Explain handshaking between sequencer and driver?
7. Explain the different types of sequencer arbitrations?
8. Why are phases used?
9. What is m_sequencer and p_sequencer?
10. What is the significance of virtual sequence and virtual sequencer?
11. Write a logic for driver in UVM?
12. Write a logic for body task of sequence.
13. Explain factory and its overriding mechanism?
14. Why is the final phase from top to bottom?
15. In which phase do we print topology?
16. What are the Methods in driver and sequencer?
17. What are reporting mechanism macros? Explain the verbosity?
18. What if we are sending the packet from monitor to scoreboard?
19. What is the issue that the packet might not be received to the scoreboard according to UVM?
20. Explain the steps, How to verify FIFO with a test case?
21. Explain UVM agent and its components?
22. What is the difference between objects and components?
23. What is Virtual Sequencer?
24. Explain various Verification methodologies?
25. What is analysis FIFO?
26. What is the start method in UVM?
27. What is 'timescale'?
28. Why is the run phase a task?
29. What is RAL?

30. What is analysis port?
31. Why is TLM FIFO used?
32. What are Reporting mechanisms?
33. What is the advantage of using UVM over other methodologies?
34. What is UVM sequencer?
35. What are the Call Backs?
36. Why is the sequencer in between driver and sequence? We are adding one extra component in the testbench?
37. How can we sample this snippet which was based on functional coverage?
38. What are the overriding methods in the UVM Factory?
39. What is the difference between new () and create ()?
40. What is the cover point?
41. What is type casting?
42. What is UVM factory overriding?
43. Explain the difference between resource_db and config_db?
44. How does the upper level component access the lower level properties in UVM?
45. Can we pass virtual interface information without uvm_config_db?
46. What is option.auto_bin_max?
47. What is a v_plan?
48. Why do we use \$value\$plusargs?
49. How is the sequence body called?
50. How to end the test case?
51. Is there any inbuilt method to do deep copy?
52. If we set any database in a test for a variable and again if we set a different value for that same variable in the environment then in the driver or in any lower components which value will we get?
53. How will you declare bidirectional ports in the interface?
54. Explain the verification flow?
55. Explain various verification methodologies?
56. What is the verification cycle?
57. Write a TB for half adder in UVM?

58. How to deal with scenarios of simulation stuck at one point in UVM?
59. How to identify the missing packets in UVM?
60. What is m_sequencer?
61. How is the monitor to scoreboard connection made?
62. How does the monitor communicate with other components?
63. Which phase is the Top down approach?
64. What is the difference between m_sequencer and p_sequencer?
65. What are UVM callbacks?
66. Explain UVM_Factory and various overriding mechanisms?
67. What is UVM? What is difference between SV and UVM?
68. Explain UVM Phases?
69. What is virtual sequencer?
70. What is resource db?
71. What are config_db methods? What are the uses of config database?
72. In which phase do we print topology?
73. What are the types of objections?
74. What is the in-built TLM port of driver and sequencer?
75. Explain the run phase of the driver?
76. How many clocking blocks are required for connecting DUT to TB?
77. Are get_next_item () and put_next_item () blocking methods?
78. What is the sum of the following verbosity levels: UVM_LOW + UVM_HIGH + UVM_NONE + UVM_DEBUG
79. Where do you write the actual stimulus: driver/ test/ sequence/ sequence_item?
80. What is the datatype of UVM_ACTIVE: structure/ string/ time/ enum/ union?
81. In which way can the driver send the response back sequence?
A) item_done (B) put_response (C) rsp_port
82. Which UVM reporting macros helps to exit the simulation?
83. How can we connect a TB component with a higher level TB component?

84. Write a generic pseudo code in UVM for master driver and master monitor components?
85. What is the difference between UVM RAL model backdoor write/read and front door write/read?
86. Explain the end of the simulation in UVM?
87. Explain the effect of grab () and lock () on sequencer?
88. What is common between UVM_do and UVM_rand sequence?
89. Why do we use sequencers?
90. How to identify the missing packets in UVM?
91. How is the sequence body called?
92. How to end the test case?
93. What does is_passive signifies?
94. Is there any inbuilt method to do deep copy?
95. What is a TLM port?
96. What are the components present in the agent?
97. Why do we connect sequence and driver via sequencer? Why not connect them directly?
98. Can we pass virtual interface information without uvm_config_db?
99. What is UVM_factory? What are the steps to create an UVM_factory?
100. What is config_db?
101. What is resource_db?
102. What are TLM ports?
103. What is an analysis port? What is the use of write() method?
104. What is the handshake mechanism between the sequence and driver?
105. What is a sequencer?
106. What is the Arbitration process of sequencer?
107. What are raise and drop objections and where we can declare them?
108. What is the flow of UVM architecture?
109. Consider, there are 8 objections raised and 7 are dropped. Whether the test file will be in run_phase or it will be end the run_phase. How to end the test?

110. Which array is usually used inside a scoreboard?
111. What are the testcases for router project?
112. Explain the difference between p_sequencer and m_sequencer?
113. Explain the difference between analysis port and TLM port?
114. Explain the approaches of various phases?
115. What are the advantages of UVM over SV?
116. What is the difference between copy and clone methods?
117. What are the arguments that has to be passed when we set the configuration?
118. Why do we need virtual sequences and sequencer? Explain it with a diagram.
119. What are the different overriding methods in factory?
120. What if objection is not raised and dropped?
121. What is the difference between UVM_component and UVM_object?
122. Write a sample monitor code in UVM?
123. What is use of factory registration?
124. What is the use of uvm_do macro?
125. What is a p_sequencer?
126. What is an analysis port and export? How we use analysis export?
127. What is the difference between set() and write() methods in RAL?
128. What is an object?
129. Differentiate between TLM and Analysis ports.
130. What is the application of uvm_create micro.
131. Which type of class the uvm_factory is ?
132. Explain the procedure to connect components in UVM.
133. What is the application of factory?
134. Explain about different types of phases and brief about the build and run phase.
135. Brief out the usage of 'uvm_create macro.
136. Explain about objections and configuration class of uvm.
137. Brief out the process between sequence, sequencer and driver.
138. Differentiate between \$finish and global stop.

139. Explain UVM Hierarchy and TLM Port connection.
140. Explain UVM sequencer method.
141. List different types of phases of UVM.
142. Explain router architecture in UVM.
143. Explain the functionalities of `get_name()` and `get_full_name()` methods in UVM.
144. Explain `get()` and `try_get()` methods of UVM.
145. Explain UVM Test bench Architecture of Router project.
146. Explain about port and export of TLM ports.
147. Write the differences between TLM Port and Analysis Port.
148. Explain the Handshaking mechanism of Sequencer and Driver.
149. Write and explain the `run_phase` for `UVM_monitor`.
150. Write a Verilog HDL code for the FSM using asynchronous active low reset and write the UVM TB to verify the same.
151. Explain the Structure of UVM Test bench.
152. Explain the Flow of UVM.
153. Write a code for UVM transaction class.
154. How to define a class in UVM?
155. How to create a class in UVM?
156. What is the difference between UVM TLM Ports & SV Mailbox?
157. Draw the UVM Architecture.
158. Why need to use `Create` method, why not `new ()` in UVM?
159. Type and instance override can be done for components or objects?
160. What is `uvm_config_db`?
161. How does the factory registration work?
162. Steps in UVM sequencer creation?
163. What are the different phases available in UVM?
164. How to use backdoor access in RAL?
165. What is the difference between `grab` and `lock`?
166. Effect of `grab ()` and `lock ()` on sequence?
167. Show the syntax for connecting a monitor and scoreboard.
168. When an agent is working in active mode, which components present?

169. When an agent is working in passive mode, which components present?
170. What is the handshaking signal between sequencer and driver?
171. What is the difference between a Component and Object?
172. What is an agent?
173. What is TLM port?
174. What is Factory in UVM?
175. What is a) TLM port b) Analysis port c) TLM FIFO
176. What is config db?
177. What is the procedure to create connection between sequencer and driver with TLM ports?
178. Explain about Active and Passive agents in UVM?
179. What are the roles of phases in UVM?
180. What are the components of UVM?
181. What is UVM_config?
182. Explain the UVM hierarchy?
183. Explain the sequence flow between sequence and driver.
184. Explain the driver to sequencer handshaking mechanisms.
185. Explain about the monitor to scoreboard connection.
186. Why do you need coverage in verification?
187. Explain verbosity based on one snippet?
188. Purpose of PLI.
189. Explain how we use Configuration in UVM.
190. What do you mean by Verification?
191. Which of the Following is not available in Verilog? Foreach, Repeat, For, Forever?
192. Write a script for how to compile and run a test case in UVM.
193. How does the scoreboard compare the monitor data with reference model data?
194. Explain the Router 1*3 project description.
195. How many Phases are there in UVM
196. Differentiate between Mailbox and TLM Ports.
197. How is Sequencer connected to driver?
198. What is difference between System Verilog and UVM Test bench?

199. What are call-backs?
200. What are the testcases for router project?
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